

CHAPTER 8

SEX-RELATED DIFFERENCES AND THEIR IMPLICATIONS FOR RESISTANCE TRAINING

N. Travis Triplett, PhD, and Nicole Dabbs, PhD

After completing this chapter, you will be able to

- discuss sex-related differences in physical characteristics and muscular function and the implications for resistance training in females;
- recognize common injuries and conditions in females;
- discuss responses and adaptations to resistance training in females; and
- understand resistance training recommendations for females.

Strength and conditioning professionals need to understand sex-related differences in physical characteristics, body composition, and physiological responses to resistance exercise when designing and evaluating resistance training programs for females. An understanding of these differences and the areas of concern that are unique to female athletes can help to optimize performance and decrease the risk of sport-related injuries.

Implications of Resistance Training for Females

Females who regularly participate in resistance training activities can improve their health, reduce their risk of degenerative diseases (e.g., osteoporosis) and their injury rates, and enhance their overall sporting performance. Whereas in the past females may have questioned the value of resistance training or even avoided this type of exercise due to social stigma, evidence clearly indicates that females are capable of tolerating and adapting to the stresses of resistance exercise and that the benefits are substantial (93). Furthermore, to enhance health and fitness and reduce injury rates, it is now suggested that resistance training be an essential component of any exercise or training program that females follow (127, 128, 167).

Physical Changes in Early Adulthood

While differences in growth and development between boys and girls are covered in [chapter 7](#), a few physical developmental changes continue into early adulthood. On average, females mature earlier than males (105), and most measures of female growth and maturation are complete by around age 18. Some exceptions include the growth plates, which typically close by the age of 20, and peak bone density (around age 30) (105). However, neither of these are undeveloped enough to result in a higher risk of injury from training, including activities such as high-intensity or high-load plyometrics (2, 122). The brain also continues to develop throughout the 20s, but most of this growth is in the frontal lobe, which controls decision-making, while the motor-related brain areas are already developmentally complete (2). The aspects of brain function that can continue to improve into the early 30s include visual-perceptual skills. Reactive agility training (see [chapter 20](#)) is a type of training that can improve these visual-perceptual skills (2). The additional

benefits beyond improvements in athletic performance due to improved reaction time and cognitive processing include a reduction in the risk of falls later in life (2).

During puberty the production of **estrogen** in girls increases fat deposition and breast development, whereas **testosterone** production in boys increases bone formation and protein synthesis. Though estrogen also stimulates bone growth, boys have a longer growth period and commence puberty at a later stage, and therefore adult men tend to achieve greater overall stature than adult women (2). On average, adult females tend to have more body fat, less muscle, and lower bone mineral density than adult males. Furthermore, females tend to be lighter in total body weight than males. Anthropometric measurements of adults indicate that males tend to have broader shoulders relative to their hips and females tend to have broader hips relative to their waist and shoulders. The broader shoulders in males can support more muscle tissue and can also provide a mechanical advantage for muscles acting at the shoulder. Although some female athletes have lower fat percentages than untrained males, extremely low body fat percentages in females may be associated with adverse health consequences (124, 125).

Maintaining skeletal muscle mass and bone mineral density is important for both athletic performance and health. Without a sustained resistance training program, muscle mass and bone density begin a pattern of decline in the mid-30s, which can accelerate after approximately age 50, especially when accompanied by menopause (120, 152). Therefore, females should be encouraged to participate in long-term resistance training programs to counteract the skeletal and muscular changes that will otherwise occur with aging. Another benefit of maintaining or increasing muscle mass aside from athletic performance is the extension of the time females can engage in sport and reduce their risk of injury (154), as well as maintaining the ability to perform activities of daily living and independence into old age (18, 121, 182).

A key focus of any resistance training program for a female should be to maintain muscle mass and bone mineral density, both of which are important for athletic performance and overall health.

Pregnancy

Pregnancy occurs only when pubertal sexual development is complete and prior to menopause (average age between 12-51 years). There are significant and specific physiological and anatomical changes that occur during each of the trimesters of pregnancy and postpartum periods, which can affect physical training and performance ([table 8.1](#)).

Multiple professional organizations have recommendations for physical activity and exercise during pregnancy (20, 21, 26, 123, 188). All of the published guidelines promote exercise as a minimal-risk activity that has benefits for most females during pregnancy, with some modifications to programming and specific exercises required as pregnancy progresses due to normal anatomic and physiologic changes that occur. These guidelines are presented in [table 8.2](#).

For pregnant females, program design variables (exercise selection, frequency, intensity, and duration) should be based on the baseline level of fitness of the athlete immediately prior to the confirmation of pregnancy. Resistance training at a moderate intensity during pregnancy has many benefits including improved blood sugar levels and improved insulin usage, as well as improved feelings of energy and reduction of fatigue (181). Studies on elite athletes have shown that they are able to manage a variety of training loads and intensities and still deliver healthy babies and return to their sport (168); however, most of the research is on more moderate loads. Vigorous-intensity exercise, including high-intensity interval training, is tolerated well among females with healthy pregnancies and who previously engaged in vigorous training, but

caution is elicited in exceeding 90% of maximum heart rate until more research is performed (13, 169).

Pregnancy is a unique condition with specific challenges to resistance training program design. An awareness of the normal physiological changes during pregnancy is very important to being able to design a safe, effective, and individualized resistance training program for a pregnant female.

TABLE 8.1 Physiological and Anatomical Changes in Pregnancy

Trimester	Cardiorespiratory	Musculoskeletal	Metabolic
First	• Increased cardiac output by 20% • Decrease in peripheral vascular resistance • Increase in blood volume • Increased tidal volume • Increase in minute ventilation (40%-50%) and $\dot{V}O_{2\max}$	• Weight gain and increased forces on joints • Increase in ligament laxity • Changes in micro-architecture of bone but not overall bone mass	• Increase in metabolic rate • Increase in body temperature
Second (in addition to prior trimesters)	• Increased cardiac output by 40% • Decrease in peripheral vascular resistance by 25%-30% • Increase in blood volume, with steady fall in hemoglobin • Fluid retention and edema • Supine positions decrease venous return to heart, which decreases stroke volume and cardiac output	• Changes in center of gravity • Increase in loss of balance	

Sources: (86a, 89b, 96a, 162a, 186a, 187a)

Trimester	Cardiorespiratory	Musculoskeletal	Metabolic
Third (in addition to prior trimesters)	• Slight decrease in cardiac output at term with stroke volume decreasing and heart rate increasing • Peripheral vascular resistance begins to increase from 32 weeks • Blood volume increases 50% of prepregnancy levels with a 50% increase in plasma volume by 34 weeks	• Increase in lumbar lordosis • Increase in foot size • Increase in compression neuropathies	
10 weeks postpartum	• Most cardiovascular and respiratory changes return to prepregnancy levels within the first 1-2 weeks after delivery	• Pelvic floor dysfunction • Urinary incontinence • Increase in ligament laxity (may persist up to 5 months after delivery)	

Sources: (86a, 89b, 96a, 162a, 186a, 187a)

TABLE 8.2 General Exercise Recommendations and Associated Signs and Symptoms of Exercising During Pregnancy

Trimester	Exercise recommendations and exercise-related signs and symptoms
First	• Continue activity at prepregnancy levels and intensities but monitor for adverse signs and symptoms. • Misinterpretation of maximum heart rate, reaching it sooner in pregnancy. Use combined approach of heart rate and perceived exertion to monitor intensity. • Initiate pelvic floor exercises to assist in labor and delivery and help with urinary incontinence. • Increase squatting activities with body weight and loaded to maintain strength and improve labor and delivery. • Increase water intake.

Sources: (86a, 89b, 96a, 162a, 186a, 187a)

Trimester	Exercise recommendations and exercise-related signs and symptoms
Second (in addition to prior trimesters)	• Avoid supine positions and exercises; switch to incline, side-lying, or vertical positions. • Continue pelvic floor exercises to assist in labor and delivery and help with urinary incontinence.
Third (in addition to prior trimesters)	• Modify activity based on signs, symptoms, and maternal comfort. • Continue to monitor for adverse signs and symptoms. • Avoid supine positions and exercises; switch to incline, side-lying, or vertical positions. • Modify activities to decrease discomfort or symptoms of neuropathy.
10 weeks postpartum	• Pelvic floor exercises (i.e., Kegel exercises) immediately after delivery for at least 12-24 weeks. • Core stabilization exercises immediately after delivery for at least 12-24 weeks. • Gradual return to physical activity and sport-related activities. • Increase water intake. • With Cesarean deliveries, more time may be needed to restore abdominal muscles to prepregnancy strength and stabilization.

Sources: (86a, 89b, 96a, 162a, 186a, 187a)

Injury and Health Considerations in Female Athletes

Female athletes face higher risks of specific injuries and health concerns than male athletes, possibly due to anatomical and biomechanical differences. Strength and conditioning professionals should emphasize lower extremity biomechanics and proper exercise techniques to address injury risk. Female athletes also experience higher rates of sport-related concussions, and mental health concerns can be notable. Sleep deprivation heightens their risk for injury and impairs recovery; thus, sleep education is essential. Additionally, pelvic floor dysfunction, especially urinary incontinence, affects female athletes at a higher rate, often affecting their activity level and risk of injury.

Anterior Cruciate Ligament

It is important for strength and conditioning professionals to be aware of **anterior cruciate ligament (ACL)** injuries in female athletes, particularly in sports such as soccer, lacrosse, and basketball (3, 4). According to a number of reports, ACL injuries occur in greater absolute frequencies in males (122), but in comparable sports female athletes' relative rate of ACL injuries is two times higher than those reported for male athletes (3, 4, 78, 86, 114). Although it is possible that the increasing absolute number of knee injuries simply reflects an increase in participation by females in organized sport, others have suggested causative factors. It is possible that the high muscular resistance and knee torque during high-load landing and rapid change-of-direction movements may cause microdamage or strain on the ACL over time, increasing the risk for injury for all athletes (12, 187). However, female athletes may have other factors contributing to an increase in the rate of ACL injury, such as anatomical, biomechanical, and external factors.

Anatomical factors, such as femoral notch size reducing tensile strength, greater tibial slope increasing strain, and ligament laxity (45, 74, 98, 163), may contribute to heightened ACL injury risk in females. However, it is believed that the most significant contributing factor is a neuromuscular deficiency, which ultimately leads to abnormal biomechanics such as increased dynamic knee valgus upon contact with the ground (128). Therefore, sociocultural factors such as motor development experience, training opportunities and resources, gendered development, and sex inequality in sporting opportunities (27, 130, 142) should be examined as causal for the higher ACL injury risk for female athletes.

Evidence suggests that in order to reduce the risk of ACL injury, youth should participate in preparatory conditioning before puberty to optimize training adaptations (128) and continue across the life span. Many ACL injuries in female athletes occur from noncontact mechanisms (e.g., deceleration, lateral pivoting or landing) (4, 23), which makes engaging in progressive overload training, plyometrics,

agility, and balance training that is designed and delivered by a strength and conditioning professional critical to reducing the risk of ACL injuries (6, 51, 79, 183). Additional clinical trials are needed to determine the best method for reducing the incidence of ACL injuries in female athletes and improving adherence to these training protocols.

Patellofemoral Pain Syndrome

Patellofemoral pain syndrome (PFPS) is a broad term used for pain arising from the patellofemoral joint, also known as *runner's knee* or *jumper's knee* (24). Female athletes are more than two times more likely to have incidences of PFPS compared to male athletes (25). PFPS has been shown to be attributed to the hip mechanics in female athletes before puberty, and modified mechanics in the knee may be related to PFPS throughout development in female athletes (60). Strength and conditioning professionals should focus on lower extremity biomechanics, exercise technique, imbalances, and weaknesses with any athlete experiencing PFPS to improve outcomes. The National Athletic Trainers' Association has a position statement that provides guidance to practitioners on how to manage patellofemoral pain in athletes, including use of bracing, taping, or other modifications (24).

Bone Stress Injuries

Bone stress injuries, including stress fractures, have regularly been shown to occur at a higher rate in female athletes than in male athletes (34, 153). The accumulation of repetitive stress, bone stress injuries frequently occur in athletes (1, 153) and are considered overuse injuries, where abnormal bone remodeling occurs (180). Strength and conditioning professionals should continue to monitor workload, and though it is challenging to identify individual bone stress tolerance, it is possible to generally monitor bone health and symptoms of **relative energy deficiency in sport (RED-S)** (124). (See [chapter 11](#) for more information on RED-S.) Additionally, other

risk factors could be identified through a screening process for bone stress injuries, such as vitamin D levels and early sport specialization (1, 180). Continuing to prescribe exercises that are osteogenically stimulating and alerting workload through periodization may assist in reducing bone stress injuries for female athletes.

Concussions

A **concussion**, or mild traumatic brain injury, is an impact to the head or body that causes the brain to move rapidly back and forth in the skull, which disrupts normal brain function (73). Female athletes are at an increased incidence of sport-related concussions in some sports (soccer, basketball, softball), with hockey having the highest incidence of concussions compared to male athletes in similar sports (35, 41, 133). Female athletes have been shown to have twice the rate of contact with equipment and surfaces compared to male athletes. However, player-to-player contact has been shown to be similar in both female and male athletes for causing concussions (32, 97). Other factors, such as biomechanical and hormonal differences, may lead to a greater susceptibility of concussions in female athletes (109, 151, 161). Female athletes experience longer recovery time with prolonged symptoms after a concussion compared to male athletes (32, 41, 109, 161). When multiple concussions occur, female athletes experience lower cerebral blood flow (70) and greater changes between single- and dual-task gait activities compared to a single concussive event (81).

Concussion management, prevention, and return-to-play programs should consider the sex differences in injury mechanics. However, due to a lack of female athlete-specific research in concussion management programs, there should be an individualized approach for concussion management and return to play based on clinical presentation, regardless of sex (109, 151). The American Medical Association for Sports Medicine position statement on concussion in sport can guide practitioners on the clinical diagnosis, assessment, and management in athletic populations (73).

How Can Female Athletes Reduce Their Injury Risk?

To help female athletes reduce their injury risk, strength and conditioning professionals should do the following:

- Recommend preparticipation screening by a sports medicine physician. This should include identification of risk factors for injury along with musculoskeletal testing.
- Encourage female athletes to participate in a year-round conditioning program that includes resistance, plyometric, speed and agility, and flexibility training. The conditioning program should meet the specific needs of each athlete and should progress in a periodized fashion that allows for training adaptations to continually manifest.
- Ensure that females learn, and can repeatedly demonstrate, correct movement mechanics (e.g., jumping, landing, twisting, and cutting) within a variety of environments.
- Precede every exercise session with a general dynamic warm-up and a specific warm-up using movements that resemble those involved in the activity and that target the activation of key muscle groups (e.g., posterior chain muscle groups).
- Provide augmented feedback within training sessions to optimize skill transfer and enhance biomechanics related to injury.

Mental Health

Mental health has gained increasingly more attention in athletes with a focus on sport organizations and practitioners to take a holistic approach when training athletes (33, 57, 77, 149). Athletes are exposed to additional mental health risk factors (77, 149), and female athletes in particular are at higher risk for certain mental health symptoms and prevalence compared to male athletes (5, 179). Research suggests that female athletes may have poor mental health outcomes due to an increased risk of certain disorders (e.g., burnout, eating disorders, disordered eating, anxiety, depression) (5, 75, 144, 149) and are subjected to increased appearance-based pressure (8, 144), discrimination, and financial strain than male athletes (179).

Research also suggests that elite female athletes are twice as likely to self-report symptoms of depression than male athletes (63), which may inhibit health and performance. In 2021, 40% of collegiate female athletes reported experiencing overwhelming anxiety compared to 22% in collegiate male athletes (189).

Early detection of mental health issues is critical, and strength and conditioning practitioners should understand symptoms and how these issues arise and should promote and provide environments that support an athlete's mental health, prevention, and treatment team and strategies of mental health disorders.

Sleep

A majority of athletes receive less-than-age-appropriate sleep levels, and a loss in sleep for athletes is a known risk factor for increased injury risk (15, 16, 157). Female collegiate athletes may be particularly likely to have low sleep durations that are consistent with sleep deprivation (<7 h/night), in association with daytime sleepiness and dysfunction (30, 69). Sleep quantity (113, 116) and quality have been shown to be lower after training and competition in female athletes (134, 136). Education on sleep hygiene for athletes, coaches, and support practitioners has been shown to address some barriers preventing good sleep quality and thus has been shown to improve sleep in athletes (115, 135). Improving sleep quality and quantity is critical for optimal recovery (178).

Pelvic Floor Dysfunction

Pelvic floor dysfunction (PFD) is a term used to identify various impairments of the pelvic floor muscles that support the pelvic organs and promote continence (99, 107). The most common PFDs are **urinary incontinence (UI)** and **stress urinary incontinence (SUI)** among several others (7, 42, 107). UI is an involuntary loss of urine, and SUI is an involuntary loss of urine with any physical

exertion (22, 150). It has been reported that UI in female athletes occurs 5 times more compared to male athletes, and 2.7 times more compared to sedentary females (42, 150, 164, 172). Female athletes have an overall 27% prevalence of UI compared to male athletes, and a 36% prevalence of UI compared to sedentary females (42, 150, 164, 172). Participating in sport has been reported to be one of the main causes of UI in female athletes, with high-impact and high-loading sports increasing incidence of SUI.

UI may be due to the imbalance of contraction strength and an increase in abdominal pressure that results in deformation of the connective tissue (99, 107). Programs should include prevention, lifestyle modifications, pelvic floor strengthening, biofeedback, and pharmacological interventions. Female athletes should be taught how to properly engage their pelvic floor muscles and abdominal wall muscles, and in some cases may need to see a physical therapist that specializes in PFD. With proper education and pelvic floor treatment, females can recover from UI (46, 55).

How Can Females Improve Their Health Outcomes?

To help females improve their health outcomes, strength and conditioning professionals should do the following:

- Recommend working with appropriate medical staff to provide resources and services as needed when they have identified signs and symptoms of concerns that increase risk of health issues
- Encourage females to seek services from a mental health provider when initial symptoms of mental health disorders arise
- Educate females on the appropriate amount and quality of sleep needed and the important role it plays in the body's recovery and for decreasing risk of injury
- Provide information to females about the variety of impairments that can occur in the pelvic floor and encourage females to seek a specialized physical therapist in PFD when signs and symptoms occur

Training Responses and Adaptations

Before resistance training programs can be properly designed, understanding the physiological responses to a resistance training session and adaptations to regularly performing a resistance training program are critical. The main body systems involved are the musculoskeletal, metabolic, and endocrine systems, which have varying levels of response and adaptation differences between males and females. These differences can affect how programming priorities are set, which is important for obtaining the optimum training results.

Musculoskeletal

Resistance training offers a multitude of benefits for females, including the attenuation of age-related declines in bone mineral density (89). Specifically, the stress from the mechanical loading applied via resistance training directly increases the magnitude of skeletal remodeling and therefore bone mass. Resistance training can increase bone mineral density in various skeletal regions of the body (89). It is also known that higher intensities of resistance training promote greater degrees of **osteogenesis** (103) and that preadolescence is an opportune time to participate in weight-bearing and other loaded activities in order to enhance bone mineral density (65, 80).

Through participation in a resistance training program, females can also experience muscle hypertrophy. When sophisticated techniques (e.g., computed tomography) are used to accurately measure changes in muscle cross-sectional area, relative short-term (up to 16 weeks) gains in muscle hypertrophy are similar between sexes (43, 67). When the muscular development of female weightlifters, bodybuilders, and track and field athletes who have not used anabolic steroids is examined, it is obvious that muscle hypertrophy is possible in females who regularly participate in high-volume or high-intensity training programs, even though these gains

may be less than in males. A genetic disposition to develop a large muscle mass may also be a contributing factor (166).

Metabolic

The metabolic system is a key system in physical activity, and understanding its function is necessary to understanding the acute responses and training adaptations of the other body systems. While the metabolic system operates similarly in males and females, some differences exist that relate to the amount and type of muscle mass, which can affect the magnitude of training-related adaptations. However, there is a lack of research comparing males and females of similar training backgrounds, as well as in high-level athletes, which limits the understanding of the metabolic system and how it adapts in female athletes.

Anaerobic Energy

The biochemical pathways that provide short-term energy include ATP-PCr and glycolysis. While all muscle fibers can produce energy with all types of biochemical pathways, the faster muscle fibers that have a greater capacity for force production have a greater amount of the substrates (phosphocreatine, glycogen) and enzymes needed for the nonoxidative biochemical pathways (see [chapter 3](#)). On average, males have a slightly higher proportion of these faster (type II) muscle fiber types (84, 165). While both males and females have the capacity to undergo muscle hypertrophy with various types of anaerobic training, as previously indicated, the level of absolute hypertrophy attained by males will exceed that of females, resulting in greater storage potential for phosphocreatine and glycogen, creatine kinase, and glycolytic enzymes, as well as greater overall glycolytic capacity (54, 155).

Aerobic Energy

The biochemical pathways that provide energy for long-duration activities involve the oxidative energy systems (i.e., Krebs's cycle,

electron transport chain, and lipid oxidation), which continuously supply energy at rest and for lower intensity activities (62). The oxidative pathways depend on the cardiorespiratory system's ability to supply oxygen and remove metabolic wastes, and research reviews have presented differences in males and females under various conditions (9, 29, 96, 186). Female endurance athletes have been shown to be more fatigue resistant during prolonged events, despite having overall lower aerobic capacity ($\text{VO}_{2\text{max}}$) relative to body size and slower performance times than males (17, 48). This is thought to be due to muscle fiber type distribution (165), with females having a greater proportion of type I oxidative fibers on average. Females therefore oxidize a greater amount of fat and a lesser amount of carbohydrate and protein (40, 170). This finding is supported by Cardinale (2018), who determined that females have close to 33% greater mitochondrial respiration than males. These differences may present an advantage to female athletes in most types of endurance activities, including ultraendurance events (174).

Training adaptations in the aerobic energy pathways differ in magnitude between males and females with regard to the cardiovascular system in particular (145), which affects oxygen delivery to the muscle tissue and subsequent oxidative energy production. In females, more modest increases in blood volume and similar or lesser increases in heart dimensions such as left ventricular chamber size and wall thickness result from aerobic endurance training (11, 44) when compared to males. However, female athletes have similar adaptations compared to males with regard to left ventricle function, including a greater ejection fraction (159). Females also present more efficient oxygen cost of running than males at various running intensities (76) while maintaining their running economy across a range of velocities. All of these factors affect overall aerobic performance, with females compensating for smaller hearts and less oxygen-carrying capacity by different mechanisms than males.

Endocrine

The **endocrine system** is responsible for regulating physiological responses and adaptations to the various stressors encountered, including physical activity. In response to strength training and conditioning, the endocrine system regulates muscular repair and growth (58, 95) as the various anabolic and catabolic mechanisms triggered by the endocrine system affect the ability of skeletal muscle to generate force and to adapt to training (95). The immediate, or acute, endocrine response to a training session traditionally has been thought to be a greater catalyst for muscle tissue growth and remodeling than the chronic changes in resting hormonal concentrations (94).

While a large number of hormones perform a myriad of functions within the human body, a much smaller number affect muscle responses to training, and even fewer have been the focus of research in athletic females. The primary hormones to have known influence on muscle and strength development, which have been studied in females, include testosterone, the growth hormone family, insulin-like growth factor-1 (IGF-1), and **cortisol**. It is also important to understand general function of the hormones involved in the **menstrual cycle**, because they can influence training and performance in female athletes and are found in much higher concentrations than in males.

Testosterone

Testosterone is present in much higher concentrations in males than in females, but females produce testosterone in small quantities in the ovaries and the adrenal gland (52). However, resting testosterone concentrations are highly variable among females, which can account for the variability in responsiveness to resistance training of absolute hypertrophy and strength (67), although relative levels of hypertrophy and strength are similar between the sexes. Despite low testosterone concentrations, females experience stimulation of muscle protein synthesis (185). With an acute bout of

resistance exercise, females do not exhibit significant elevations in testosterone concentrations (95, 177). However, with prolonged training, some acute elevations in testosterone have been observed in females (137). These increases are likely due to a combination of factors including a decreased rate of degradation, evidenced by lower measures of metabolites of testosterone breakdown (49). Therefore, in females, testosterone likely still has an important role in the development of absolute muscular size, strength, and power with resistance training and may also be an important indicator of the trainability of an individual (67, 68).

Growth Hormones and Growth Factors

Growth hormones (GH) have a primary role in muscle tissue remodeling and repair but do not function alone, because muscle protein synthesis is triggered with the **growth factor** known as IGF-1. In females, it has been demonstrated that resting concentrations of GH are higher than in males (92) and with chronic training, GH levels at rest are lower in females compared to males (171), which could indicate a greater sensitivity to the hormone, but research is lacking.

After chronic resistance training, circulating GH rises similarly in females and males. Following an acute resistance exercise bout in females, circulating GH concentrations increase two- to fourfold (66, 83). The magnitude of the circulating GH increase depends on the type of exercise stimulus as well as the amount of total work performed. For example, exercise sessions consisting of high total work (10RM load) and shorter rest intervals (1 minute) (92), large muscle groups (95), multi-set protocols (126), and exercises that induce elevations in blood lactate and reductions in blood glucose (91, 95) result in significant elevations in GH. With training, the magnitude of the exercise-induced rise in GH increases (171).

Cortisol

Cortisol is an important hormone that helps regulate blood glucose levels as well as inflammation. Cortisol rises in anticipation of exercise and sport competition in order to prepare the body for the upcoming psychological and metabolic demands (118, 140). This anticipatory rise has been shown to be greater in trained females than in sedentary females (47). The cortisol response to acute exercise is greatest in resistance exercise protocols that result in higher levels of GH (91) and blood lactate (90). Several investigations have demonstrated that in comparison to males, the cortisol response to acute exercise is attenuated in females (58, 66, 140). This is thought to be due to higher cortisol sensitivity in females (156) or the impact of female sex hormones on modulating activity of the sympathetic nervous system, adrenal glands, and hypothalamus-pituitary axis (175).

Training influences the magnitude of the cortisol response in females. Elite female athletes do not exhibit an anticipatory rise in cortisol before exercise (47) but do exhibit higher cortisol secretion and different patterns of secretion (158). In addition, female sex hormones (**estradiol** and **progesterone**) may lower cortisol responsivity, thereby potentially altering the effects of cortisol over the menstrual cycle (104). Immune factors modulate glucocorticoid sensitivity (59), accounting for differences observed in female athletes.

The Menstrual Cycle

Females experience the menstrual cycle during adolescence and adulthood, approximately between the ages of 12 and 50 (56). While the cycle is usually discussed in terms of reproductive function, the menstrual cycle hormones also influence other body functions that are related to exercise and training (37). There are also times during the menstrual cycle when the body experiences symptoms such as cramping, nausea, and headaches, which may negatively affect the ability to engage in exercise sessions or a

training program. Therefore, it is important that practitioners have a general understanding of the menstrual cycle and its physical effects.

In brief, two primary hormones are released from the ovaries during the menstrual cycle, estradiol and progesterone. During the initial half of the cycle, menstrual bleeding occurs and then estradiol peaks and drops. Progesterone rises and peaks during the second half of the cycle, and there is a second but lesser peak of estradiol. By the end of the cycle, both estradiol and progesterone drop to their lowest levels ([figure 8.1](#)).

Estradiol

In menstruating adult females, the primary form of estrogen is estradiol, which has been shown to directly affect the structure and function of muscle, tendon, and ligaments, as well as contribute to muscle mass and strength (37). Estradiol appears to have a more protective effect on skeletal muscle (100) by attenuating the amount of damage. To do this, estradiol may decrease postexercise membrane disruption, act as an antioxidant, act as a membrane stabilizer, and mediate inflammatory cell infiltration (53, 141). In addition, estradiol may influence contractile properties of skeletal muscle and muscle protein turnover (71). This supports muscle mass development by promoting protein synthesis over degradation and enhancing sensitivity of the response to resistance training (71, 72). High levels of estradiol are not always beneficial because decreases in power and performance can result, which can increase the risk for catastrophic ligament injuries (37). Low levels of estradiol are equally problematic, resulting in increased muscle cell death and reduced skeletal muscle mass and strength (39, 87).

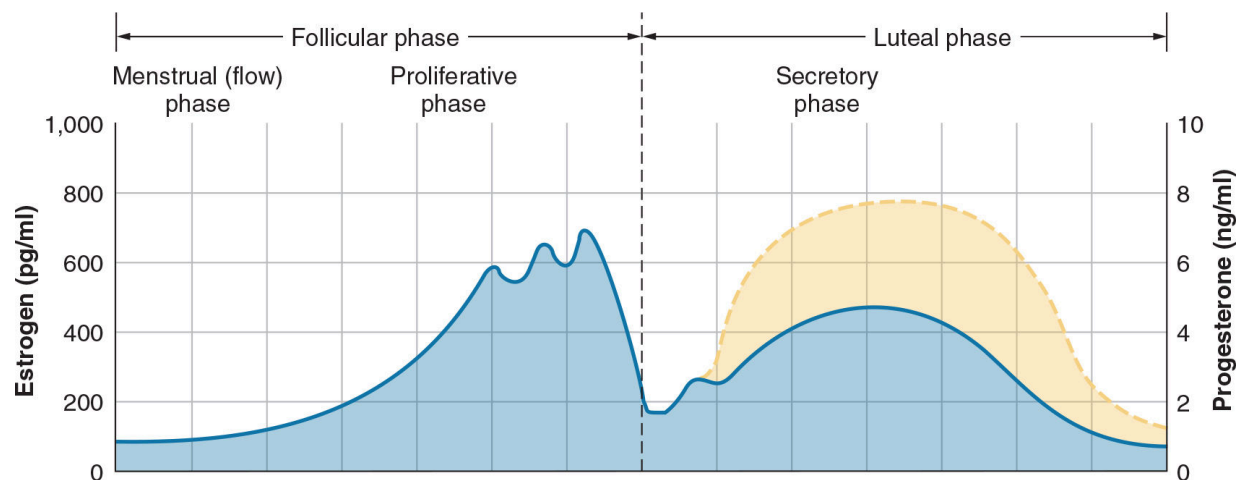


FIGURE 8.1 Phases of the menstrual cycle and the concomitant changes in estrogen and progesterone.

Reprinted by permission from W.L. Kenney, J.H. Wilmore, and D.L. Costill, "Sex Differences in Sport and Exercise," in *Physiology of Sport and Exercise*, 9th ed. (Human Kinetics, 2024).

After an acute bout of resistance exercise, increased estradiol concentrations have been reported (94, 158). This response may occur due to increased activity of the enzyme that converts testosterone to estrogen (estradiol) (129), which may also contribute to increases in circulating testosterone (101). Because of the beneficial effects of estrogen on muscle, it is important for females to maintain normal menstrual cycle function in order to maximize their adaptations to training (37) and reduce the risks associated with low estradiol levels (173).

Resistance Training Considerations

With resistance training, the primary goal is to improve strength and power. When comparing training-induced changes in muscular strength between sexes, it is important to distinguish between absolute and relative measures. Sex-related differences in hypertrophy and muscle mass (106, 132, 143) indicate that absolute muscle strength in females is lower than in males (108, 117), whereas males' greater absolute strength is primarily achieved by larger muscle fiber size (117, 146, 160). The absolute lower body strength of females is generally closer to male values compared to the absolute values for upper body strength (10, 82, 84, 117). Sex-related differences in body composition, anthropometric characteristics, and distribution of fat-free mass can partly explain these sex-related differences, which are apparent in recreationally trained individuals as well as highly trained athletes (14).

When considered on a relative basis, sex-related differences in muscular strength are greatly reduced. Females have similar number of muscle fibers as males, and strength is similar when normalized to muscle volume (113, 117). Differences in hypertrophy between males and females may be due to a variety of factors such as endocrine and genetics because they relate to regulating muscle mass (184), whereas factors such as muscle cross-sectional area and enzyme activity may contribute to strength and performance differences (85, 131, 162). Because the average male and female differ considerably in body size, it is useful to compare sex differences in strength relative to body weight, fat-free mass, and muscle cross-sectional area (14, 31). When expressed relative to body weight, the lower body strength of females is similar to that of males, while the upper body strength of females is still somewhat less (10, 82, 84, 117). If comparisons are made relative to fat-free mass, differences in strength between males and females tend to disappear (10, 82, 84, 117).

Sex-related differences in power output are similar to those for muscular strength. Measurements comparing power outputs of female athletes relative to lean body mass were reported to be similar for lower body power and still presented differences for upper body power when compared to male athletes (10, 61). Although the data are equivocal, sex-related differences in the rate of force development (147, 148) and the recruitment differences in muscle activation where females express lower levels of type IIa fibers' percent of myosin heavy chain isoform content (138, 139) could partly explain these findings.

During progressive overload and targeted training, female athletes experience substantial power and strength increases similar to male athletes (64). In a recent study, maximal strength and vertical jump height were increased in collegiate volleyball players by using a combination of traditional resistance training exercises and weightlifting variations at various loads in addition to volleyball practice, over the course of about 2.5 years of training (88). Other research supports plyometrics in female basketball players and other velocity-based resistance training in cyclists to improve strength, power, and muscle hypertrophy (36, 119). It has also been reported that adding heavy strength training to a female athlete's endurance training increased muscle strength, performance, and size without affecting endurance running performance or running economy (176).

Several studies have investigated menstrual cycle phase and exercise performance, with some effects indicating that the phase of cycle has an effect on performance in eumenorrheic females (111). Additional research shows that in naturally cycling females, there are minimal differences in strength measures between phases (19). The practical relevance of phase-based training comes into question of whether it is feasible to track female athletes' cycles when considering the variety of hormonal contraceptives they may be using or menstrual cycle irregularities, especially in team-based sports in which the number of athletes is high.

The most recent evidence on menstrual cycle and performance from meta-analyses did not conclude any systematic relationship between performance and menstrual cycle phase in trained female athletes (38, 50, 111, 112). Other considerations are necessary to consider, such as varying menstrual cycle symptoms among female athletes and the effect on their sleep and recovery during all stages of the menstrual cycle (110). Consequently, the current evidence does not support modifying prescription of resistance training based on a female athlete's menstrual cycle phase.

Conclusion

Evidence indicates that resistance exercise can be a safe and effective method of training for female athletes of all ages and abilities. The potential benefits are multifactorial, including positive effects on a variety of physical performance variables (e.g., strength and power), health markers (e.g., body composition and cardiac function), and psychosocial development (e.g., self-image and confidence). Moreover, regular participation in resistance training programs can better prepare female athletes to increase performance and decrease risk of injury. Though the fundamental principles of resistance training are similar for people of both sexes, there are unique considerations specific to female athletes. Knowledge of sex-related differences is essential to the development and evaluation of safe and effective resistance training programs. Strength and conditioning professionals should be aware that individual responses to resistance exercise may vary greatly independent of sex and should be sensitive to the individual needs of all participants.

Over the past few decades, coaches, clinicians, and exercise scientists have added to our understanding of sex-related differences and their implications for resistance exercise. Their work has quantified the impact of resistance training on female athletes and has provided the foundation for recommendations about the design of strength and conditioning programs. The information in this chapter and other chapters should help strength and conditioning professionals understand and appreciate sex-related differences and enhance their ability to develop safe and effective resistance training programs for females.

KEY TERMS

anterior cruciate ligament (ACL)
bone stress injuries

concussion
cortisol
endocrine system
estradiol
estrogen
growth factor
growth hormones (GH)
menstrual cycle
mental health
osteogenesis
patellofemoral pain syndrome (PFPS)
pelvic floor dysfunction (PFD)
pregnancy
progesterone
relative energy deficiency in sport (RED-S)
stress urinary incontinence (SUI)
testosterone
urinary incontinence (UI)

STUDY QUESTIONS

1. Which of the following is not one of the physical changes related to growth and development that occur well into adulthood and may affect resistance training and conditioning in females?
 - a. increased bone mineral density
 - b. closure of growth plates
 - c. improved visual-perceptual skills
 - d. increased muscle mass
2. Which of the following musculoskeletal changes occurs with pregnancy and can affect resistance training programming?
 - a. increased body temperature
 - b. decreased cardiac output
 - c. increased ligament laxity

- d. decreased venous return in supine positions
3. Estradiol, one of the main hormones in the menstrual cycle, has which of the following effects in the body?
 - a. promotes muscle protein synthesis and reduces muscle protein degradation
 - b. increases the amount of muscle damage with training
 - c. negatively influences the contractile properties of muscle
 - d. increases postexercise muscle cell membrane disruption
 4. When a female athlete has multiple concussions, which of the following does she not experience?
 - a. lower cerebral blood flow
 - b. fewer changes between single-task gait activities
 - c. greater changes between single-task gait activities
 - d. greater changes between dual-task gait activities
 5. When creating a program for a female athlete, which of the following is the most important to consider?
 - a. individualization
 - b. sex differences
 - c. sport differences
 - d. generalization by sport